§2 Formal definition of an ACS — part 4 of 5

Reference: subsec:state_space_basic_variables, subsec:viability_function_and_blockage

Source: paper/section2_definition.tex, lines 2132–3131 (§2.7–§2.8), at commit e6e12cd. The Japanese manuscript is authoritative. Parts 1–3 are 02_definition_1.md, 02_definition_2.md and 02_definition_3.md.

2.7 State transition, action and external-environment input

§2.7 describes the discrete-time dynamics of an ACS. It fixes the one-step state evolution by the state-transition map f whose type was declared in def:acs(4), and introduces the basic assumptions used to describe the dynamics. Some of these assumptions are forward-referenced from the basic type declarations of §2.4. The subsection also treats the decomposition of f into an autonomous component that does not pass through action selection and a directed component mediated by action selection, and the dynamical non-degeneracy with respect to external-environment input.

The semantics of the required environment space E was given in §2.6. The state-transition map treated here is defined on all of E_{ext}, and describes the state evolution under one and the same type whether or not the required environmental factors are satisfied.

New symbols: the autonomous process f_{auto}, the directed process f_{dir}, the combination operation $\oplus$, and the one-step difference ΔY_t. The subscript auto stands for *autonomous* and dir for *directional*. "Autonomous" here means that a component of the state transition does not depend on action selection. It is a different concept from *automaticity* treated in §4, which concerns the mode of operation of the mechanism and has no correspondence with f_{auto}. The symbol $\oplus$ is a binary operation combining two state values and does not denote addition in a group or ring.

Compactness and continuity

Assumption (ass:compact_spaces): compactness of X and U. The state space $X \subseteq \mathbb{R}^n$ and the action space $U \subseteq \mathbb{R}^m$ are compact (closed and bounded). *Introduction status: S (structural). Strength: strong.* The assumption provides the structural basis for dealing with the compactness of closed subsets of X and U, the existence of extrema of continuous real-valued functions, and the compactness of the feasible action set A_t (prop:At_compact). Standard facts on compactness, closed sets and continuous maps follow the usual results of general topology (cited in the manuscript as Munkres 2000). The assumption is not imposed on the external environment state space E_{ext}. Relaxation to non-compact or locally compact spaces is a Phase 2 matter.

Assumption (ass:f_continuity): continuity of f. The state-transition map $f : X \times U \times E_{\mathrm{ext}} \to X$ is continuous. *Introduction status: S. Strength: medium.* It is a structural assumption to keep closedness and limit operations manageable; together with the closedness of the feasible set K it is used to derive the closedness of the feasible action set A_t (prop:At_compact). Continuity is stated with respect to the topology of E_{ext} (§2.2, subsec:wellformedness). Since no continuity or other regularity is imposed on a required environmental factor η (def:acs(3)), the assumption does not stipulate openness, closedness or other topological properties of $E_{\mathrm{req}}(x)$.

Assumption (ass:Eext_time_fixed): time-fixedness of E_{ext}. In Phase 1 the external environment state space $E_{\mathrm{ext},t}$ is treated as one and the same topological space E_{ext} independently of time. *Introduction status: S. Strength: medium.* It simplifies the deterministic discrete-time framework of Phase 1. It does not exclude the realized external environment state e_t changing with time. Extension to a time-dependent external environment state space is a Phase 2 matter.

The state-transition map

Definition (`def:state_transition`): state transition. The state of an ACS individual Σ_i evolves in discrete time by

$$x_{t+1} = f(x_t, u_t, e_t),$$

where $x_t \in X$ is the state vector, $u_t \in U$ the effective action, and $e_t \in E_{\text{ext}}$ the realized external environment state. This is consistent with the standard notation of discrete-time control systems (Bertsekas 2017). For a real- or vector-valued time-dependent quantity Y_t, the one-step difference is $\Delta Y_t := Y_{t+1} - Y_t$. This notation is not applied to set-valued, categorical or relational quantities for which subtraction is undefined; the state space X itself is a set and is not the object of a time difference.

The discrete-time formulation is conceptually consistent with the idea of feasible sets and feasible actions in Aubin's viability theory (Aubin 1991). "Consistent" here does not mean mathematical equivalence or inclusion; it means the absence of conceptual contradiction in describing state transition centered on feasibility-preserving constraints. Extension to continuous time is a Phase 2 matter.

Reading convention for f (`rem:f_reading`). The paper reads $u \in U$ as the model's action input, and $f(x, u, e)$ as the next state determined by internal state x, action input u and external environment state e. Under this reading, an attempt at an action, a failure to execute it, and the degree of success of execution are not distinguished inside the description of f. Phase 1 introduces no independent variable for success or degree of execution; if such differences are to be included in a model, they are absorbed into the state x, the external environment state e, the way action inputs are taken, or the definition of f. For fixed (x, u, e), f gives one next state. The effective action u_t of `def:acs`(2) likewise does not mean that execution actually succeeded; it is the model's action input fed to f after the internal command has been resolved. No structure expressing executability of an action input separately from U is introduced in Phase 1. Distinctions not expressed by this convention are treated as theoretical limits in §9.

Definition (`def:env_realization`): transition map of the action-response component, and environment realization. Under the decomposition of `def:Eext_decomposition`, a total, single-valued map

$$g_O : O \times X \times U \times E^{\text{bg}} \longrightarrow O$$

is called the **transition map of the action-response component**, and the realized values of the action-response component evolve by $o_{t+1} = g_O(o_t, x_t, u_t, b_t)$. The a-th component of $g_O(o, x, u, b)$ is written $g_O(o, x, u, b)_a$. The sequence $(b_t)_{t\geq 0}$ of the exogenous background component is given exogenously. The quadruple $(E^{\text{bg}}, O, g_O, (b_t)_{t\geq 0})$ is called the **environment realization** of the ACS. g_O is not a component of the ACS 7-tuple; it is a structure on the environment side used in analysis. The ACS is not required to hold or compute g_O. The paper imposes no continuity or other regularity on g_O.

From `def:state_transition` and `def:env_realization`, given an initial value (x_0, o_0), a background sequence $(b_t)_{t\geq 0}$ and an effective-action sequence $(u_t)_{t\geq 0}$, the sequence $(x_t, o_t)_{t\geq 0}$ is uniquely determined by

$$x_{t+1} = f(x_t, u_t, (b_t, o_t)), \qquad o_{t+1} = g_O(o_t, x_t, u_t, b_t).$$

When $\mathcal{A}_O = \emptyset$, g_O is a constant map to a one-point set and the realized external environment state sequence is determined by the background sequence alone; this coincides with the case in which the whole external-environment-state sequence is given exogenously.

On the generation of realized external environment states (`rem:exogenous_environment`). `def:state_transition` does not place the generation rule for the sequence $(e_t)_{t\geq 0}$ inside the 7-tuple. In Phase 1 only the background sequence $(b_t)_{t\geq 0}$ is given exogenously; the action-response component may respond to the ACS's own actions according to g_O. Hence the trajectory of an ACS is determined not by the 7-tuple Σ alone but relative to Σ together with an environment realization. This relativity holds equally when the whole external-environment-state sequence is given exogenously. A generation rule for the background sequence, coupled dynamics treating other ACS individuals as individuals with their own transition maps, and stochastic environments are Phase 2 matters (L2 in §9). Describing the behavior of other ACS individuals as an action-response component inside g_O is a model on the environment side

and does not give those individuals' own transition maps. The relational structure between the ACS and external objects is treated in §6; §6 gives a description structure of relations and does not give transition rules for external environment states.

Definition (`def:action_difference`): action difference (classification C). For $(o, x, b) \in O \times X \times E^{\mathrm{bg}}$, an action $u \in U$ and a slot $a \in \mathcal{A}_O$, if

$$g_O(o, x, u, b)_a \neq g_O(o, x, u_\emptyset, b)_a,$$

then u is said to have an **action difference** at (o, x, b) in slot a. Here $u_\emptyset$ is selective waiting (`rem:action_space_structure`). Immediately from the definition, selective waiting $u_\emptyset$ has no action difference in any slot at any (o, x, b); and when $\mathcal{A}_O = \emptyset$ the action difference is not defined.

Scope of the action difference (`rem:action_difference_scope`). The action difference expresses only the presence or absence of a one-step difference in value relative to waiting. Five limitations apply.

(a) It is relative to the choice of component O_a (which distinctions are expressed as values); a difference that the values of O_a do not distinguish does not appear, so absence of an action difference does not mean absence of physical contact or influence.
(b) The reference of comparison is fixed at $u_\emptyset$; differences relative to other actions are not its object.
(c) The magnitude (distance) of an action difference is not specified.
(d) It is not a judgment of causal contribution; cases of overdetermination or pre-emption are not distinguished (§9).
(e) The paper places no general implication between $u \in U_{\mathrm{ext}}$ and having an action difference; an externally directed action having no action difference because of distance, bodily conditions or other circumstances is not excluded.

Representation of intake into the self side (`rem:intake_joint_transition`). Intake of an external thing or quantity into the self side is not represented as one and the same mathematical object moving across types. It is represented as a pair of two transitions within the same one step: for some slot a, $o_{t,a} \neq \bot_a$ and $g_O(o_t, x_t, u_t, b_t)_a \neq o_{t,a}$ (including the case where the value becomes $\bot_a$), together with the fact that $f(x_t, u_t, e_t)$ can depend on $o_{t,a}$. The type-separation convention TSC is maintained, and the identity of a material is not tracked across the types X and O. From the viewpoint of a required environmental factor η, one can read that before intake $\eta(x_t, e_t) = 1$ holds through the external-side state, and after intake satisfaction of the same factor holds through the internal state x_{t+1} (`rem:env_utilization_function`). The remark introduces no new map, variable, constitutive requirement or assumption. A classification of types of intake, and the balance of resource quantities common to outside and inside, are not treated in Phase 1.

Two-component decomposition of the state transition

Assumption (`ass:f_decomposition`): two-component decomposition of the state transition. The state-transition map f treated in Phase 1 is expressed, by an autonomous functional process f_{auto}, a directed process f_{dir} and a continuous binary operation $\oplus$, as

$$f(x_t, u_t, e_t) = \oplus(f_{\mathrm{auto}}(x_t, e_t),\ f_{\mathrm{dir}}(x_t, u_t, e_t)),$$

with $f_{\mathrm{auto}} : X \times E_{\mathrm{ext}} \to X$, $f_{\mathrm{dir}} : X \times U \times E_{\mathrm{ext}} \to X$, and a continuous binary operation $\oplus : X \times X \to X$. f_{auto} does not depend on action selection and represents the autonomous component determined by internal state and external environment state; f_{dir} takes the effective action u_t as an argument and represents the directed component that can depend on it. The assumption does not require that f_{dir} actually depends on u_t, i.e. that some $u \neq u'$ give $f_{\mathrm{dir}}(x, u, e) \neq f_{\mathrm{dir}}(x, u', e)$. Further, the autonomous process is not a constant map: there exist $(x, e), (x', e') \in X \times E_{\mathrm{ext}}$ with $f_{\mathrm{auto}}(x, e) \neq f_{\mathrm{auto}}(x', e')$. For selective waiting $u_\emptyset$ (`rem:action_space_structure`), for every $x \in X$ and $e \in E_{\mathrm{ext}}$,

$$\oplus(f_{\mathrm{auto}}(x, e),\ f_{\mathrm{dir}}(x, u_\emptyset, e)) = f_{\mathrm{auto}}(x, e).$$

Introduction status: S. Strength: strong. Since a general continuous f need not have this decomposition, it is stated explicitly as a structural assumption of Phase 1. It serves to separate, in the mathematical description, the autonomous component that does not pass through

action selection from the directed component mediated by action selection, and to describe selective waiting as $f(x, u_\emptyset, e) = f_{\text{auto}}(x, e)$. However, the assumption does not stipulate that, in a blocked state, the implementation-level autonomous functional process corresponding to f_{auto} actually keeps operating, or that the effective time horizon $\hat{\tau}_t > 0$ holds (`rem:blockage_auto`). Nor does it require $\mathrm{WF}_{i,t} = 1$ to be maintained while f_{auto} operates. Concrete implementation forms of f_{auto}, f_{dir} and $\oplus$ are Phase 2 and later matters.

Algebraic status of $\oplus$ and selective waiting (`rem:oplus_neutrality`). `ass:f_decomposition` does not require general algebraic structure of $\oplus$ such as associativity, commutativity or a global identity element. What is needed of $u_\emptyset$ is only the displayed identity for each (x, e); $f_{\text{dir}}(x, u_\emptyset, e)$ is not required to be a global identity element of $\oplus$. The neutrality of $u_\emptyset$ is thus a local condition on the value of f_{auto}.

Non-degeneracy with respect to external-environment input

Assumption (`ass:env_input_nondegeneracy`): dynamical non-degeneracy with respect to external-environment input. f is non-degenerate with respect to external-environment input: there exist $x \in X$, $u \in U$ and $e \neq e'$ in E_{ext} with $f(x, u, e) \neq f(x, u, e')$. *Introduction status: S. Strength: weak.* It excludes from Phase 1 degenerate systems in which differences of external environment state have no effect on the state transition. It does not say that environmental differences affect feasibility or the feasible action set; non-degeneracy at the level of the feasible action set is treated in §2.8.

Levels of non-degeneracy (`rem:nondegeneracy_levels`). The paper distinguishes three levels of non-degeneracy with respect to external-environment input. The first is the state-transition level of `ass:env_input_nondegeneracy` (environmental differences can affect the next state). The second is the level of the feasible action set, treated in §2.8 (environmental differences can affect the set of feasible actions). The third is the level of sensing, treated in §2.9 (`subsec:three_primitive_functions`; environmental differences can affect sensing output). Non-degeneracy at the feasible-action-set level implies `ass:env_input_nondegeneracy` (`prop:nondegeneracy_implication`); the converse does not hold in general (`rem:nondegeneracy_converse`). For the sensing level, unless a connecting condition is separately placed, the paper does not assume mutual implication with the other two levels. None of the three is the same condition as the constitutive requirement CR-1 of the required environment space E (§2.6). The paper does not state the existence of required environmental factors as a general consequence of these non-degeneracies, nor these non-degeneracies as a general consequence of CR-1.

2.8 Persistence evaluation, feasibility and one-step environmental feasibility

§2.8 defines the persistence evaluation function Π for the feasibility of a state, and from it derives the feasible set K and the feasible actions. These feasibility notions connect with the idea of control-invariant sets in control theory (Blanchini 1999). A state with no feasible action is defined as **blockage**, and the set of external environment states allowing one-step feasibility is derived.

Collapse is an event distinct from blockage; its definition was given in §2.5 (`def:collapse`). The subsection makes explicit that the paper does not state, as a general consequence of blockage, either failure of well-formed realization ($\mathrm{WF}_{i,t} = 0$, `def:wf_realization`) or collapse.

The feasibility notions of §2.8 are a different object from the satisfaction of required environmental factors (§2.6). The required-environment satisfaction set $E_{\text{req}}(x)$ is determined by satisfaction of required factors alone, whereas the one-step feasible environment correspondence defined here is determined by the existence of a feasible action. Their relation is organized in `rem:fixed_environment_and_state_dependent_viability`.

New symbols: the persistence evaluation function Π, the threshold vector θ, the feasible set K, the feasible action correspondence $A(x, e)$, the feasible action set A_t at time t, and the one-step feasible environment correspondence $\mathcal{E}^{(1)}_{\text{viab}}(x)$. The superscript (1) is an identifier

for “one step”, not a power. The symbol $\rightrightarrows$ denotes a set-valued correspondence.

Persistence evaluation function and feasible set

Definition (`def:Pi`): persistence evaluation function. Given a threshold vector $\theta = (\theta_1, \dots, \theta_n) \in \mathbb{R}^n$, the **persistence evaluation function** $\Pi : X \to \mathbb{R}$ is

$$\Pi(x) := \min_{k \in \{1,\dots,n\}} (x_k - \theta_k), \qquad x \in X.$$

x_k is the k-th component of x and θ_k the feasibility threshold of that dimension; $\Pi(x)$ gives the smallest of the per-dimension margins $x_k - \theta_k$ (evaluation by the weakest dimension; `rem:min_type_pi`). Applied to the state at time t: $\Pi(x_t) = \min_k (x_{k,t} - \theta_k)$. In Phase 1 each θ_k is a time-invariant constant; extension to dynamic thresholds is a Phase 2 or later matter.

Why a min-type Π (`rem:min_type_pi`). If each θ_k is defined as the lowest feasible level of its dimension, i.e. a level below which feasibility is impaired regardless of the other dimensions, then feasibility requires $x_k \geq \theta_k$ for all k simultaneously. Under this non-compensatory threshold convention, $\Pi = \min_k (x_k - \theta_k)$ evaluates feasibility by the dimension with the smallest margin and directly represents the lowest feasible levels. Forms allowing compensation between dimensions, such as weighted-linear or some norm types, are not consistent with the non-compensatory meaning of each θ_k adopted here; so Phase 1 adopts the min type as the basic form. This does not exclude other continuous-function representations that give the same K as a level set at 0 or above (`prop:theta_from_pi`; under stated assumptions).

Definition (`def:K`): feasible set. From Π the **feasible set** is constructed as the derived definition

$$K := \{x \in X \mid \Pi(x) \geq 0\}.$$

K is a closed subset of X (by continuity of Π and closedness of $\{x \in \mathbb{R} \mid x \geq 0\}$).

Proposition (`prop:theta_from_pi`): equivalence of feasibility and derivability of K. (i) Under `def:Pi`, for all $x \in X$: $\Pi(x) \geq 0 \iff x \in K$. (ii) Under $K \neq \emptyset$ and closedness of K, independently of the defining form of Π, K can be represented as a level set at 0 or above of a suitable continuous function $\Pi : X \to \mathbb{R}$. Statement (ii) shows only that a nonempty closed set K is representable as such a level set; it is not claimed that the function can be written in the min form of `def:Pi`. *Proof outline.* (i) $\min_k (x_k - \theta_k) \geq 0$ iff $x_k \geq \theta_k$ for all k iff $x \in K$. (ii) For $K \neq \emptyset$ the distance function $\mathrm{dist}(\cdot, K)$ is real-valued and 1-Lipschitz, hence continuous, so $\Pi_d := -\mathrm{dist}(\cdot, K)$ is continuous; by closedness of K, $\mathrm{dist}(x, K) = 0$ iff $x \in K$, so $K = \{x \mid \Pi_d(x) \geq 0\}$.

Feasible action set and blockage

Definition (`def:at`): feasible action correspondence and feasible action set. For $x \in X$ and $e \in E_{\mathrm{ext}}$, the **feasible action correspondence** $A : X \times E_{\mathrm{ext}} \rightrightarrows U$ is

$$A(x, e) := \{u \in U \mid f(x, u, e) \in K\},$$

and the **feasible action set** at time t is $A_t := A(x_t, e_t)$. $A(x, e)$ is a time-independent correspondence and A_t is its realized value at time t. When the individual is made explicit, with the components of Σ_i: $A_i : X_i \times E_{\mathrm{ext},i} \rightrightarrows U_i$, $A_i(x, e) := \{u \in U_i \mid \Pi_i(f_i(x, u, e)) \geq 0\}$ and $A_{i,t} := A_i(x_{i,t}, e_{i,t})$ (the description-level convention of §2.4). For a single ACS, $A_i = A$ and $A_{i,t} = A_t$. $A(x, e)$ is the set of actions that put the one-step next state into K; in particular, for $x \in K$ it is the set of actions that keep the next state within K. Read as a set-valued correspondence it connects with set-valued analysis (Aubin and Frankowska 1990).

Reading of the feasible action set (`rem:at_reading`). `def:at` is not changed by this remark. The paper reads $A(x, e)$ as the **set of structural response potentials** at that (x, e): the set of action inputs that put the next state into K. This reading is a descriptive name. Neither the reading nor `def:at` includes as a requirement that an element of $A(x, e)$ is actually selected, or that feasibility is maintained over several time steps. Selective waiting $u_\emptyset$ is also an element of U, and if $f(x, u_\emptyset, e) \in K$ then $u_\emptyset \in A(x, e)$. The relation between action selection and the feasible action set is treated in §3.

Proposition (`prop:At_compact`): closedness and compactness of the feasible action set. Under `ass:compact_spaces` (compactness of U) and `ass:f_continuity`, for every $x \in X$ and $e \in E_{\text{ext}}$, the value $A(x, e)$ is a closed, hence compact, subset of U (including the case $A(x, e) = \emptyset$).
Proof outline. $u \mapsto f(x, u, e)$ is continuous and K is closed (`def:K`), so $A(x, e)$ is the preimage of a closed set under a continuous map and is closed in U; a closed subset of the compact U is compact; the empty set is compact.

Definition (`def:blockage_sec2`): blockage. When the feasible action set is empty at time t, $A_{i,t} = \emptyset$, the ACS individual Σ_i is said to be in a **blockage** state. In blockage, whatever $u \in U$ is chosen, the one-step next state cannot be put into K: $\forall u \in U,\ f(x_t, u, e_t) \notin K$.

Blockage is not failure of well-formed realization or collapse (`rem:blockage_not_collapse_sec2`). $A_{i,t} = \emptyset$ is an event distinct from collapse. The paper uses "collapse" only where $\text{Col}_{i,t} = 1$ holds, i.e. where, by time t, a state has come about in which the organizational relations that established the identity as a structural individual have been lost in a way that does not recover as the same individual (`def:collapse`). Blockage, by contrast, expresses that, under the current state and realized external environment state, no action exists that puts the next state into K. So the paper does not state collapse $\text{Col}_{i,t} = 1$ as a general consequence of blockage, nor failure of well-formed realization $\text{WF}_{i,t} = 0$ at that time (`def:wf_realization`). Even under blockage, $\text{WF}_{i,t} = 1$ is possible as long as well-formed realization as the same ACS individual is kept (`rem:time_identity_collapse`). The distinction between blockage and collapse, and the organization of the routes to each, are treated in §7.

Temporal self-identity versus functional termination, collapse and blockage (`rem:time_identity_collapse`). The preliminary condition (TI-1) states that comparison of temporal self-identity presupposes well-formed realization at both times. The paper does not state functional termination or collapse as a general consequence of $\text{WF}_{i,t} = 0$ alone, nor of blockage $A_{i,t} = \emptyset$. Times at which blockage holds and $\text{WF}_{i,t} = 1$ are not excluded, and (TI-1) can hold there. The relation of functional termination and collapse to well-formed realization follows `rem:collapse_wf_relation`.

Figure (`fig:s2_termination`). The logical relations among blockage, failure of well-formed realization, functional termination and collapse are shown in a two-part figure. Panel (a) shows implications at the same time t: collapse $\text{Col}_{i,t} = 1$ implies functional termination $\text{FT}_{i,t} = 1$ (`prop:termination_order`(ii)) and failure of well-formed realization $\text{WF}_{i,t} = 0$ (`prop:termination_order`(iv)); no line is drawn between blockage and the other three, and where no line is drawn the figure displays no implication (`rem:blockage_not_collapse_sec2`, `rem:termination_hierarchy`, `rem:collapse_wf_relation`). Panel (b) shows, for the case in which collapse holds at some time, the time-wise indicators: $\text{FT}_{i,t}$ and $\text{Col}_{i,t}$ are history indicators and, once 1, remain 1 (`prop:collapse_basic`); the figure draws the case where the functional-termination state first holds before the time t_C at which the collapse state first holds, and includes the case where both first hold at the same time (`prop:termination_order`(i)(ii)); at times after t_C the functional-termination state is not judged (`prop:termination_order`(iii)); the values after collapse are displayed by the history indicator and the attribution provision and do not represent continuation of a realization trajectory of the same individual (`def:collapse`); the figure does not fix the values of $\text{WF}_{i,t}$ except after t_C (= 0). Blockage is not placed as a stage on the time axis.

Describability of state transition under blockage (`rem:blockage_auto`). Blockage means that, whatever action is chosen, the one-step next state cannot be put into K. It does not mean that f or its autonomous component f_{auto} becomes mathematically undefined; $f(x_t, u, e_t)$ stays defined on the domain of f for every $u \in U$, though its next state is not in K. If $\text{WF}_{i,t} = 1$ is maintained and the realization conditions needed for the state transition are maintained, the state change after blockage can also be described by $x_{t+1} = f(x_t, u_t, e_t)$. However, the paper does not state, as a general consequence of the definition of blockage or of `ass:f_decomposition` alone, that the implementation-level autonomous functional process corresponding to f_{auto} actually keeps operating during blockage, or that the effective time horizon $\hat{\tau}_t > 0$ holds, and it gives no conditions for these. $\hat{\tau}_t > 0$ is equivalent to the prediction-accuracy coefficient being positive (§5, `def:tau_hat_section5`), and its value depends on the readout rule specified by the describer.

Scope of the meaning of blockage (`rem:blockage_scope_sec2`). $A_{i,t} = \emptyset$ expresses only that, under the current state and realized external environment state, no action exists that places the next one-step state in K.

(a) Blockage at a time with $x_t \in K$ is read as **blockage in the maintenance phase** (leaving K at the next time cannot be avoided by any action); blockage at a time with $x_t \notin K$ is read as **blockage in the recovery phase** (return to K is not possible in one step). These are names for readings; no definition or symbol is added (cf. the maintenance and recovery phases of §3).
(b) The paper does not state, as a general consequence of blockage in the recovery phase, that return to K in finitely many steps is impossible; an individual in the middle of a recovery requiring several steps may be in recovery-phase blockage at each time until the step just before the recovery completes.
(c) The feasible action set and blockage are relative to the adopted one-step transition specification, i.e. the time resolution and the correspondingly fixed action space U and transition map f; the paper does not assume that the judgment of blockage is preserved between formalizations with different one-step specifications.
(d) The paper does not state $A_{i,t} = \emptyset$ as a general consequence of any of the following: physical immobility, stalled purposes or tasks, unavoidable loss of feasibility several steps ahead, the ACS finding no effective action candidate internally, or the ACS evaluating that it has no move. Their coexistence with blockage at the same time is not excluded.

One-step environmental feasibility

Definition (`def:one_step_viable_environment`): one-step feasible environment correspondence (classification C). For each state $x \in X$, the correspondence $\mathcal{E}^{(1)}_{\mathrm{viab}} : X \rightrightarrows E_{\mathrm{ext}}$ is

$$\mathcal{E}^{(1)}_{\mathrm{viab}}(x) := \{e \in E_{\mathrm{ext}} \mid A(x,e) \neq \emptyset\},$$

the set of external environment states that leave at least one one-step feasible action at state x. It is derived from f, K, U and E_{ext} and adds no independent component to the 7-tuple.

One-step feasibility versus required-environment satisfaction (`rem:fixed_environment_and_state_dependent_viability`). Within one Phase 1 analysis E_{ext} is fixed (`ass:Eext_time_fixed`), while $\mathcal{E}^{(1)}_{\mathrm{viab}}(x)$ can change with the internal state x; a change in the range of environments the ACS can cope with as its internal state changes is thus expressed as a change of $\mathcal{E}^{(1)}_{\mathrm{viab}}(x)$ inside a fixed E_{ext}. $\mathcal{E}^{(1)}_{\mathrm{viab}}(x)$ is not a set whose elements are each individually necessary; it is the set of external environment states permitting one-step feasibility. It is determined differently from $E_{\mathrm{req}}(x)$ (`def:E_req`), and the paper requires no inclusion in either direction between them: both the case where no feasible action exists under an external environment state in which all required factors are satisfied, and the case where a one-step feasible action exists under one in which required factors are not satisfied, are not excluded. Also, the paper does not state, as a general consequence of one-step feasibility itself, feasibility over several time points, periodic environmental demands, or reachability of future environment states; describing periodic or delayed environmental demands requires a separate formulation of environment sequences and state trajectories over several time points. For the realized state e_t: $e_t \notin \mathcal{E}^{(1)}_{\mathrm{viab}}(x_t) \iff A(x_t, e_t) = \emptyset$. Blockage then holds (`def:blockage_sec2`), but is distinguished from collapse $\mathrm{Col}_{i,t} = 1$ (`def:collapse`).

Realized external environment state versus internal estimate (`rem:actual_environment_and_internal_estimate`). $e_t \in E_{\mathrm{ext}}$ in §2.8 is the realized external environment state in the system description. It is distinguished from the environment estimate formed by the internal model, and agreement between the two is not assumed. The type of the internal environment estimate and its relation to the realized external environment state are fixed in §2.10 (`subsec:internal_model_M`). Since no metric, pseudo-metric or estimation-error evaluation map on E_{ext} is introduced here, the discrepancy between them is not quantified.

Assumption (`ass:env_viable_nondegeneracy`): non-degeneracy of the feasible action set with respect to environment input. Differences of external environment state can affect the feasible action set: there exist $x \in X$ and $e \neq e'$ in E_{ext} with $A(x,e) \neq A(x,e')$. *Introduction status: S. Strength: weak.* It excludes from Phase 1 degenerate systems in which environmental differences have no effect at all on the set of feasible actions. It does not say that such a difference originates in the satisfaction of required environmental factors.

Proposition (`prop:nondegeneracy_implication`): implication between levels of non-degeneracy. `ass:env_viable_nondegeneracy` implies `ass:env_input_nondegeneracy`. *Proof outline.* Take x and $e \neq e'$ with $A(x,e) \neq A(x,e')$ and, without loss of generality, $u \in A(x,e) \setminus A(x,e')$. By `def:at`, $f(x,u,e) \in K$ and $f(x,u,e') \notin K$; since membership in K differs, $f(x,u,e) \neq f(x,u,e')$, which gives a tuple satisfying `ass:env_input_nondegeneracy`.

The converse (`rem:nondegeneracy_converse`). The converse of `prop:nondegeneracy_implication` does not hold in general: even if $f(x,u,e) \neq f(x,u,e')$, when both belong to K or both lie outside K, the feasibility of u does not change and $A(x,e) = A(x,e')$ can result. Hence the paper does not state non-degeneracy at the feasible-action-set level as a general consequence of non-degeneracy at the state-transition level. The relation to non-degeneracy at the sensing level is organized in §2.9 (`subsec:three_primitive_functions`); unless a connecting condition is separately placed, the paper assumes no mutual implication between the sensing level and the other two levels.

Assumption (`ass:viable_realizability`): feasible realizability. Under an external environment state in which the required environmental factors are satisfied, at least one feasibility-preserving one-step transition exists: there exist $x \in K$, $e \in E_{\text{req}}(x)$ and $u \in U$ with $f(x,u,e) \in K$. *Introduction status: S. Strength: medium.* It excludes from Phase 1 degenerate systems in which no feasibility-preserving transition exists at any state and required-environment-satisfied state. It does not require $e_t \in E_{\text{req}}(x_t)$ at each time on a state-transition sequence.

Consequences of feasible realizability (`rem:viable_realizability_consequences`). From `ass:viable_realizability`: first, since there exists $x \in K$, $K \neq \emptyset$, so the hypothesis $K \neq \emptyset$ placed in `prop:theta_from_pi`(ii) is satisfied under this assumption. Second, for that x and e, $u \in A(x,e)$, hence $A(x,e) \neq \emptyset$ and $e \in \mathcal{E}^{(1)}_{\text{viab}}(x) \cap E_{\text{req}}(x)$; so there is at least one state at which the required-environment satisfaction set and the one-step feasible environment correspondence have a nonempty intersection. Third, the existence of x with $E_{\text{req}}(x) \neq \emptyset$ required by (CR-1b) (§2.6) follows from this assumption within $x \in K$. However, the paper does not state as a general consequence that the whole of CR-1 follows from this assumption, or that this assumption follows from CR-1: CR-1 is a constitutive requirement placing the 7-tuple structure in the class of ACS, whereas this assumption limits the range in which later propositions hold.

References

Aubin, Jean-Pierre. 1991. *Viability Theory*. Birkhäuser.
Aubin, Jean-Pierre, and Hélène Frankowska. 1990. *Set-Valued Analysis*. Birkhäuser.
Bertsekas, Dimitri P. 2017. *Dynamic Programming and Optimal Control, Volume i*. 4th ed. Athena Scientific.
Blanchini, Franco. 1999. “Set Invariance in Control.” *Automatica* 35 (11): 1747–67.
Munkres, James R. 2000. *Topology*. 2nd ed. Prentice Hall.